DISPROOF OF EVENTUAL EQUALITY IN ERDŐS PROBLEM 878 (QUESTION 3B)

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For $n \in \mathbb{N}$, let $\mathcal{P}(n)$ denote the set of prime factors of n (so $\mathcal{P}(0) = \emptyset$), and for a prime p let $q_p(n) := p^{\lfloor \log_p n \rfloor}$ (with $\lfloor \log_p 0 \rfloor := 0$), which for $1 \le n$ is the largest power of p not exceeding n. Erdős [1] defined

$$f(n) := \sum_{p \in \mathcal{P}(n)} q_p(n), \qquad F(n) := \max \sum_{a \in A} a,$$

where $A \subseteq [2, n]$ ranges over finite sets of pairwise coprime integers a with $\mathcal{P}(a) \subseteq \mathcal{P}(n)$, and asked whether the maxima $\operatorname{maxUpTo} f\, x := \max_{n \le x} f(n)$ and $\operatorname{maxUpTo} F\, x := \max_{n \le x} F(n)$ satisfy $\operatorname{maxUpTo} f\, x = \operatorname{maxUpTo} F\, x$ for all sufficiently large x.

Theorem. *For every $X_0 \in \mathbb{N}$, there exists $x \ge X_0$ such that* $\operatorname{maxUpTo} f\, x < \operatorname{maxUpTo} F\, x$.

Consequently, it is not true that $\operatorname{maxUpTo} f\, x = \operatorname{maxUpTo} F\, x$ for all sufficiently large x, which answers Question 3b negatively.

Correspondence with the Lean formalization. The proof below is the one formalized in `FormalConjectures/ErdosProblems/878.lean`, step by step, with the same names ($q_1, \dots, q_6$, Q_6, C_6, N_{smooth}, N_2, A, M, step, rem, K, Y_0, Y_a, Y_b, Q_a, Q_b, R', D, r_a, r_b, b_i, J, ℓ, z_j, …) and with every inequality in the form used in Lean. All variables are natural numbers, except $\operatorname{rem}(k)$, which is real. The remaining correspondences are:

This note	Lean
$\mathcal{P}(n)$; $\lfloor \log_p n \rfloor$; $q_p(n)$	`n.primeFactors`; `Nat.log p n`; `q_p p n`
$\lfloor a/b \rfloor$; $[a, b]$; $\{0, \dots, M-1\}$	`a / b` in $\mathbb{N}$; `Finset.Icc a b`; `Finset.range M`
$\operatorname{maxUpTo} f\, x$	`maxUpTo f x` (maximum over $n \in \{0, \dots, x\}$)
an integer $a \in [2, n]$ with $\mathcal{P}(a) \subseteq P$	`IsAdmissibleTerm n P a`
$9 q_p(\lfloor 19Y/18 \rfloor) < Y$	`IsGoodForPrime p Y`
$q_1, \dots, q_6$ are six distinct primes; $0 < C_6$	`Q6_prime`, `Q6_card`; `C6_pos`
Theorem; the sentence after it	`erdos_878_q3_disproved`; `erdos_878.parts.iii_eventually`
Lemma 1	`lemma1_exists_smooth_gt`; the choice of j is `exists_step_crossing`
inequality (1)	`exists_F_ge_of_smooth`, via `q_p_admissible` and `sum_le_F_of_disjoint`
Lemma 2	`lemma2_superlinear_maxUpTo_f`
Lemma 3	`lemma3_exists_good_Y`
contradiction hypothesis	`h_no_jump`
Step 1	`single_step_jump`; the decomposition of $f(n)$ is `sum_sdiff_erase_two_le`, and $r_a r_b \prod_{p \in R'} p \le n$ is `mul_mul_prod_erase_le`
Step 2: iterating Step 1; rem; $18(z + \operatorname{rem}(k)) \le 19Y \Rightarrow z \le \lfloor 19Y/18 \rfloor$; the induction on k	`iter`; `rem`; `le_cap`; `h_seq`
monotonicity of q_p and maxUpTo; $g(n) \le \operatorname{maxUpTo} g\, x$ for $n \le x$ $(g = f, F)$	`q_p_mono`, `maxUpTo_mono`; `le_maxUpTo`

The Lean file states Step 1 as the separate lemma `single_step_jump` before Lemmas 2 and 3; its hypotheses beyond those of Step 1 (inequality (1), the contradiction hypothesis for $x \ge Y$,

$18 \le Y$, $N_{\mathrm{smooth}} \le Y$, and the conclusion of Lemma 3 for Y) are the standing assumptions of Step 1 here.

Lemma 1. *If $p < r$ are primes with $9r < 10p$, then for every n with $p^{p^2+2} \le n$ there exists an integer $z \in [2, n]$ with $\mathcal{P}(z) \subseteq \{p, r\}$ and $9n < 10z$.*

Proof. Put $J = p^2 + 1$ and $\ell = \lfloor \log_p n \rfloor$; then $J < \ell$. Let $z_j = p^{\ell-j} r^j$. Then $z_0 \le n$, as $z_0 = q_p(n)$, while

$$n < p^{\ell+1} = p^{\ell-J}(p^J p) \le p^{\ell-J}(p+1)^J \le z_J,$$

since $p^J p \le p^J + J p^{J-1} \le (p+1)^J$ (the first inequality because $J = p^2+1$). Choose $j < J$ with $z_j \le n$ and $n < z_{j+1}$. Since $z_j r = z_{j+1} p$, we have $9np < 9z_{j+1}p = 9z_j r < 10 z_j p$, so $9n < 10 z_j$; as $2 \le n$, also $2 \le z_j$. Hence $z = z_j$ has the required properties. □

Fix the six primes

$$q_1 := 10000019, \quad q_2 := 10000079, \quad q_3 := 10000103,$$
$$q_4 := 10000121, \quad q_5 := 10000139, \quad q_6 := 10000141,$$

let $Q_6 := \{q_1, \dots, q_6\}$, let $C_6 := \prod_{p \in Q_6} p$, and let $N_{\mathrm{smooth}} := q_1^{q_1^2+2} + q_3^{q_3^2+2} + q_5^{q_5^2+2}$. For every m with $N_{\mathrm{smooth}} \le m$, Lemma 1 applies to the pairs (q_1, q_2), (q_3, q_4), and (q_5, q_6) (as $q_{2i-1} < q_{2i}$, $9q_{2i} < 10 q_{2i-1}$, and $q_{2i-1}^{q_{2i-1}^2+2} \le N_{\mathrm{smooth}} \le m$) and gives $b_1, b_2, b_3 \in [2, m]$ with $\mathcal{P}(b_i) \subseteq \{q_{2i-1}, q_{2i}\}$ and $9m < 10 b_i$. If moreover $R' \subseteq \mathcal{P}(m)$, $R' \cap Q_6 = \emptyset$, and $Q_6 \subseteq \mathcal{P}(m)$, then b_1, b_2, b_3 and $q_p(m) \in [2, m]$ ($p \in R'$) are pairwise coprime with prime factors dividing m, so $b_1 + b_2 + b_3 + \sum_{p \in R'} q_p(m) \le F(m)$ and hence

$$27m + 10 \sum_{p \in R'} q_p(m) < 10 F(m). \tag{1}$$

Lemma 2. *For every $R \in \mathbb{N}$, there is N_2 such that $Ry < \mathrm{maxUpTo}\, f\, y$ for all $y \ge N_2$.*

Proof. Since $\sum_p p^{-1} = \infty$, choose N with $2R + 1 \le \sum_{p<N} p^{-1}$, and set $P = \prod_{p<N} p$ and $N_2 = 2P$. For $y \ge N_2$, let $m = P\lfloor y/P \rfloor$; then $m \le y$ and $y < 2m$. Every prime $p < N$ divides m and satisfies $mp^{-1} \le q_p(m)$, so

$$(2R+1)m \le m \sum_{p<N} p^{-1} \le \sum_{p<N} q_p(m) \le \sum_{p \in \mathcal{P}(m)} q_p(m) = f(m) \le \mathrm{maxUpTo}\, f\, y,$$

and $y < 2m$ gives $Ry < \mathrm{maxUpTo}\, f\, y$. □

Lemma 3. *For every $X_0 \in \mathbb{N}$, there is Y with $\max(X_0, 18) \le Y$ and $9q_p(\lfloor 19Y/18 \rfloor) < Y$ for all $p \in Q_6$.*

Proof. Let $Y_0 = \max(X_0, 18)$, and for $p \in Q_6$ let $\mathrm{bad}(p)$ be the set of $m < 7$ for which $9q_p(\lfloor 19 Y_0 10^m/18 \rfloor) < Y_0 10^m$ fails. We show $|\mathrm{bad}(p)| \le 1$. Let $m_1 < m_2$ lie in $\mathrm{bad}(p)$, and put $Y_a = Y_0 10^{m_1}$, $Y_b = Y_0 10^{m_2}$, $Q_a = q_p(\lfloor 19 Y_a/18 \rfloor)$, and $Q_b = q_p(\lfloor 19 Y_b/18 \rfloor)$. Then $10000000 \le p$, $18 \le Y_a$, $10 Y_a \le Y_b$, $Y_b \le 1000000\, Y_a$, $Q_a \le \lfloor 19 Y_a/18 \rfloor$ and $Q_b \le \lfloor 19 Y_b/18 \rfloor$ (so $18 Q_a \le 19 Y_a$ and $18 Q_b \le 19 Y_b$), $Y_a \le 9 Q_a$, and $Y_b \le 9 Q_b$. Hence $18 Q_a \le 19 Y_a < 20 Y_a \le 2 Y_b \le 18 Q_b$, so $Q_a < Q_b$, and since Q_a and Q_b are powers of p, $Q_a p \le Q_b$. Therefore

$$20000000\, Y_a = 2 \cdot 10000000\, Y_a \le 18\, p\, Q_a \le 18\, Q_b \le 19\, Y_b \le 19 \cdot 1000000\, Y_a = 19000000\, Y_a,$$

which contradicts $18 \le Y_a$. Thus $\sum_{p \in Q_6} |\mathrm{bad}(p)| \le 6 < 7$, so some $m < 7$ lies in no $\mathrm{bad}(p)$, and $Y = Y_0 10^m$ satisfies $\max(X_0, 18) \le Y$ and $9q_p(\lfloor 19Y/18 \rfloor) < Y$ for all $p \in Q_6$. □

Proof of the Theorem. Suppose for contradiction that $\mathrm{maxUpTo}\, F\, x \le \mathrm{maxUpTo}\, f\, x$ for all $x \ge X_0$. Put $A := 1200 C_6 + 1$. By Lemma 2 with $R = 2(A+10)$, there is N_2 such that $2(A+10)y < \mathrm{maxUpTo}\, f\, y$ for all $y \ge N_2$. Let Y be given by Lemma 3 applied to $\max(X_0, N_2, N_{\mathrm{smooth}})$; then $X_0 \le Y$, $N_2 \le Y$, $18 \le Y$, $N_{\mathrm{smooth}} \le Y$, and $9q_p(\lfloor 19Y/18 \rfloor) < Y$ for all $p \in Q_6$.

Step 1. Suppose $Y \le y$, $y \le \lfloor 19Y/18 \rfloor$, $1 \le M$, and $19(M+8)Y \le 18\, \mathrm{maxUpTo}\, f\, y$. Choose $n \le y$ with $\mathrm{maxUpTo}\, f\, y = f(n)$.

- $0 < n$: otherwise $\operatorname{maxUpTo} f\, y = f(0) = 0$, contradicting $19(M+8)Y \le 18 \operatorname{maxUpTo} f\, y$ and $18 \le Y$.
- Since $f(n) \le n\,|\mathcal{P}(n)|$ and $18n \le 19Y$, we have $19(M+8)Y \le 18f(n) \le 18n\,|\mathcal{P}(n)| \le 19Y\,|\mathcal{P}(n)|$, so $M + 8 \le |\mathcal{P}(n)|$.
- Since $|\mathcal{P}(n)| - |Q_6| \le |\mathcal{P}(n) \setminus Q_6|$, $|\mathcal{P}(n) \setminus Q_6| - M \le |(\mathcal{P}(n) \setminus Q_6) \setminus \{0, \dots, M-1\}|$, $|Q_6| = 6$, and $M + 8 \le |\mathcal{P}(n)|$, we get $1 < |(\mathcal{P}(n) \setminus Q_6) \setminus \{0, \dots, M-1\}|$. Take distinct r_a, r_b in this set, so that $r_a, r_b \in \mathcal{P}(n) \setminus Q_6$, $M \le r_a$, and $M \le r_b$.
- Put $R' := (\mathcal{P}(n) \setminus Q_6) \setminus \{r_a, r_b\}$ and $D := \big(\prod_{p \in R'} p\big) C_6$. Since $r_a r_b \prod_{p \in R'} p$ divides n, we have $r_a r_b \prod_{p \in R'} p \le n$; together with $M^2 \prod_{p \in R'} p \le r_a r_b \prod_{p \in R'} p$ and $n \le 2Y$ (as $n \le y \le \lfloor 19Y/18 \rfloor$), this gives $DM^2 \le 2C_6 Y$, that is,

$$D \le \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor.$$

- Let $m := D(\lfloor y/D \rfloor + 1)$; then $y < m$ and $m \le y + D$. Since $D \mid m$, we have $R' \subseteq \mathcal{P}(m)$ and $Q_6 \subseteq \mathcal{P}(m)$; also $R' \cap Q_6 = \emptyset$ and $N_{\text{smooth}} \le m$, as $N_{\text{smooth}} \le Y \le y < m$. So (1) gives $27m + 10 \sum_{p \in R'} q_p(m) < 10F(m)$. Next, since $n \le y < m$, $\sum_{p \in R'} q_p(n) \le \sum_{p \in R'} q_p(m)$, and since every prime factor of n lies in $\{r_a, r_b\} \cup R' \cup Q_6$,

$$f(n) \le q_{r_a}(n) + q_{r_b}(n) + \sum_{p \in R'} q_p(n) + \sum_{q \in Q_6} q_q(n),$$

where $q_{r_a}(n) \le n \le y$ and $q_{r_b}(n) \le n \le y$. Each $q \in Q_6$ satisfies $9q_q(n) \le 9q_q(\lfloor 19Y/18 \rfloor) < Y$; summing over the six elements of Q_6 gives $9 \sum_{q \in Q_6} q_q(n) < 6Y \le 6y$. Hence $3f(n) < 8y + 3 \sum_{p \in R'} q_p(m)$, and, as $Y \le y < m$,

$$30f(n) + Y \le 30f(n) + y < 81m + 30 \sum_{p \in R'} q_p(m) < 30F(m).$$

Finally, $F(m) \le \operatorname{maxUpTo} F\,(y + D) \le \operatorname{maxUpTo} f\,(y + D)$, by $m \le y + D$ and the contradiction hypothesis (as $X_0 \le Y \le y + D$). Hence $y' := y + D$ satisfies

$$y \le y', \qquad y' \le y + \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor, \qquad 30 \operatorname{maxUpTo} f\, y + Y \le 30 \operatorname{maxUpTo} f\, y'. \tag{2}$$

Step 2. Iterating Step 1, that is (2), gives: if $Y \le y$, $1 \le M$, $19(M+8)Y \le 18 \operatorname{maxUpTo} f\, y$, and $y + j\lfloor 2C_6 Y/M^2 \rfloor \le \lfloor 19Y/18 \rfloor$, then there is $y' \ge y$ with $y' \le y + j\lfloor 2C_6 Y/M^2 \rfloor$ and $30 \operatorname{maxUpTo} f\, y + jY \le 30 \operatorname{maxUpTo} f\, y'$. This follows by induction on j: for $j = 0$ take $y' = y$; for $j + 1$, the y' obtained for j satisfies $Y \le y'$, $y' \le y + j\lfloor 2C_6 Y/M^2 \rfloor \le \lfloor 19Y/18 \rfloor$, and $19(M+8)Y \le 18 \operatorname{maxUpTo} f\, y \le 18 \operatorname{maxUpTo} f\, y'$, so Step 1 applies to it.

Let $\operatorname{rem}(k) := \frac{64C_6 Y}{A+k}$, and note that $18(z + \operatorname{rem}(k)) \le 19Y$ implies $z \le \lfloor 19Y/18 \rfloor$, since $0 \le \operatorname{rem}(k)$ and $z \in \mathbb{N}$. We construct $y_k \ge Y$ by induction on k with

$$18(y_k + \operatorname{rem}(k)) \le 19Y \quad \text{and} \quad 19(A + 1 + k + 8)Y \le 18 \operatorname{maxUpTo} f\, y_k.$$

For $k = 0$, take $y_0 = Y$: we have $18 \operatorname{rem}(0) \le Y$ because $18 \cdot 64C_6 Y \le (1200C_6 + 1)Y$, so $18(Y + \operatorname{rem}(0)) \le 19Y$; and $19(A + 1 + 0 + 8)Y \le 36(A + 10)Y < 18 \operatorname{maxUpTo} f\, Y$ because $2(A + 10)Y < \operatorname{maxUpTo} f\, Y$. Given y_k, set $M = A + 1 + k$, so that $1 \le M$, and step $= \lfloor 2C_6 Y/M^2 \rfloor$. Since step $\cdot\, M^2 \le 2C_6 Y$ and $(A + k)(A + k + 1) \le M^2$, we have $32 \cdot \text{step} \cdot (A + k)(A+k+1) \le 64C_6 Y$, hence $32 \cdot \text{step} \le \operatorname{rem}(k) - \operatorname{rem}(k+1)$ and $18(y_k + 32 \cdot \text{step} + \operatorname{rem}(k+1)) \le 18(y_k + \operatorname{rem}(k)) \le 19Y$, so $y_k + 32 \cdot \text{step} \le \lfloor 19Y/18 \rfloor$. Iterating Step 1 with $j = 32$ (possible as $Y \le y_k$, $1 \le M$, and $19(M+8)Y = 19(A + 1 + k + 8)Y \le 18 \operatorname{maxUpTo} f\, y_k$) gives $y' \ge y_k$ with $y' \le y_k + 32 \cdot \text{step}$ and $30 \operatorname{maxUpTo} f\, y_k + 32Y \le 30 \operatorname{maxUpTo} f\, y'$. Put $y_{k+1} := y'$. Then $Y \le y_{k+1}$, $18(y_{k+1} + \operatorname{rem}(k+1)) \le 19Y$, and

$$\begin{aligned} 95(A + 1 + (k+1) + 8)Y &\le 95(A + 1 + k + 8)Y + 96Y \\ &\le 90 \operatorname{maxUpTo} f\, y_k + 96Y \le 90 \operatorname{maxUpTo} f\, y_{k+1}. \end{aligned}$$

Finally, let $K := \big(\operatorname{maxUpTo} f \lfloor 19Y/18 \rfloor\big) + 1$. Since $18(y_K + \operatorname{rem}(K)) \leq 19Y$, we have $y_K \leq \lfloor 19Y/18 \rfloor$, so $\operatorname{maxUpTo} f\, y_K \leq \operatorname{maxUpTo} f \lfloor 19Y/18 \rfloor < K$. But $18K \leq 19(A + 1 + K + 8)Y \leq 18 \operatorname{maxUpTo} f\, y_K$, a contradiction. □

References

[1] P. Erdős, *On two unconventional number theoretic functions and on some related problems*, Topics in classical number theory, Vol. I, II (Budapest, 1981), Colloq. Math. Soc. János Bolyai **34** (1984), 445–458.